Electronic supplementary material

Units matter: cooperation in large language model agents tracks how the payoff matrix is written, not what it means

This document holds the per-model and per-cell breakdowns behind the summary statistics of the main text, the validation of the strategy read-out, and the open-weight corpus of the earlier preprint, which the main text no longer uses. Tables S1–S10 are generated directly from the machine-readable analysis output by `Analysis/scripts/42_supp_tables.py`; none of their numbers is transcribed by hand, and re-running the pipeline regenerates this document and the main text together.

S1 Validating the strategy read-out

The read-out described in §2.4 of the main text assigns each agent-game a label from the four canonical memory-one rules in two stages: an enumeration that proves whether a trajectory is exactly what a rule would have played, and where no rule fits, a long short-term memory network [1] that returns the nearest one. This section records what each stage is worth, because the strategy results of §3.10 rest on it and because two thirds of the labels come from the learned stage.

Validation corpus. The network is trained and evaluated on synthetic play generated from the same four rules, at execution-noise levels of 0 and 5 per cent, with a separate network for each of the two horizons the project uses. Ten per cent and twenty per cent noise are held out entirely, and are reported below because real transcripts are noisier than the training distribution and it matters how the network degrades outside it. Every game in the corpus of the main text runs ten rounds, so the ten-round network reads all of it; the thirty-round network is used only for the open-weight corpus of §S2.

Accuracy. At ten rounds the pipeline recovers the generating rule with accuracy 0.995 on noise-free play and 0.975 at five per cent noise, against Bayes-optimal single-label ceilings of 0.996 and 0.987. At thirty rounds the corresponding figures are 0.953 at five per cent noise against a ceiling of 0.966. Where exactly one rule fires the label is right 99.5 per cent of the time at ten rounds and 98.7 per cent at thirty, which is what makes the provable stage worth running first even though it covers a minority of games.

The identifiability floor. On noise-free thirty-round play the forced single-label ceiling is 0.625, and this is a property of the game rather than a failure of the method: an agent that cooperates throughout against a cooperator is exactly consistent with AllC, tit-for-tat and win-stay lose-shift at once, and no procedure can separate them. Allowing the first stage to return the *set* of rules that fit, the answer is correct on 100 per cent of that file with a mean answer size of 2.13 labels. This is why the main text records such runs as *ambiguous* rather than resolving them, and why the ambiguous share is treated throughout as a statement about identifiability rather than about strategy.

Where the corpus sits relative to the training distribution. The exact-rule coverage the read-out achieves on the real corpus locates it against the synthetic noise levels. Over the corpus of the main text 19.4 per cent of agent-games carry a provable rule: 17.2 per cent over the three models that met ten scales and 26.7 per cent over the three that met three. Both figures sit below the 39.1 per cent of ten-round synthetic play at ten per cent noise and, for the ten-scale group, close to the 14.3 per cent at twenty per cent noise. Frontier transcripts therefore depart from the canonical rules by more than the training corpus does, and the network is extrapolating there more often. On those two held-out noise levels its accuracy where no rule fits is 0.908 and 0.788, which brackets what these labels are worth and is the reason §3.10 of the main text reports every test on the four named rules as well as on all five categories. Table S7 gives the full provenance breakdown, scale by scale.

S2 The open-weight corpus of the earlier preprint

An earlier preprint by the present authors reported that the cooperation rate of small open-weight models varies with the payoff scale and with the prompt language [2]. That corpus is not part of the analysis in the main text, is not pooled with it anywhere, and no claim in the main text rests on it. We report it here for two reasons: it is where two of the preprint's conclusions came from, and both are conclusions the frontier sweep revises, so a reader who knows the preprint should be able to see exactly where the two corpora part company.

The corpus. Gemma 3 12B, Llama 3.1 8B and Qwen3 8B each met six payoff scales, $\lambda \in \{0.01, 0.1, 1, 10, 100, 1000\}$, the same five prompt languages and the same four ordered persona pairings, over thirty rounds rather than ten. Every model × language × scale cell holds 80 agent-games, giving 3,600 dyads, 7,200 agent-games and 216,000 decisions. The base payoff matrix, the FAIRGAME templates and the notation regimes are those of §2 of the main text. The horizon differs, so this corpus is read with the thirty-round network of §S1 rather than the ten-round one, and no level statistic here is comparable with a frontier number.

What replicates. All three models fail the invariance. The largest shift in cooperation between two payoff scales is 0.316 for Gemma 3 12B, 0.198 for Llama 3.1 8B and 0.267 for Qwen3 8B, against dyad-level permutation nulls of 0.121, 0.047 and 0.095, each at $P < 0.001$. The smallest of those three swings exceeds the median spread between the three models at a fixed scale, 0.114, and 8 of the 45 model-pair by scale-pair comparisons change sign somewhere in the sweep. That is the central result of the main text, obtained on a disjoint set of models at a different horizon, and it is the part of the preprint the denser frontier sweep strengthens rather than qualifies.

What does not: the notation account. Pooled over this arm, the three-level notation regime is the best specification in the description ladder, ahead of a specification with no λ term by 373.5 BIC points, a linear term in $\log_{10} \lambda$ by 247.2, a quadratic by 109.6, the fractional flag by 56.1 and the specification saturated in λ by 19.6; it is also the best specification for Llama 3.1 8B and Qwen3 8B individually, with Gemma 3 12B preferring a cubic in $\log_{10} \lambda$. On a six-point decade grid, however, the regime read from the printed cells and a step function of λ are the same variable: every scale that prints with a decimal point is also a small-stakes scale, with no exception anywhere on the grid. The ten-scale frontier grid breaks that coincidence at $\lambda = 0.5$, whose cells print as integers although $\lambda < 1$, and once it is broken no notation specification is the best one for any model (§3.2 of the main text). The preprint's conclusion was what its grid could support; it is not what a grid that can separate the two accounts supports.

What does not: the coordination invariant. In this arm the rate of miscoordination, the share of rounds on which the two agents chose differently, swings by 0.039 across the six scales and does not clear its null ($P = 0.14$, null 95th percentile 0.047), while mutual cooperation over the same rows swings 0.198, mutual defection 0.233 and the cooperation rate itself 0.216. Read alone, that is an invariance. On the ten-scale frontier grid the same measure does clear its null, at 0.076 against 0.056 ($P = 0.0005$). What survives across both corpora is an ordering and not an invariance: miscoordination is the smallest mover of the six measures in both, and it is fixed in neither.

The persona and the strategy mix. The persona result is qualitatively the same here, and table S11 gives it per model. All three models move toward compliance across the fractional-to-unit boundary, by +0.244, +0.052 and +0.192, and all three move back away from it into the large regime, by −0.205, −0.049 and −0.058, the first two of which exclude zero. The second half of that pattern is what the frontier corpus does not reproduce: there the three shifts out of the unit regime are −0.034, −0.033 and +0.018, and none excludes zero. All three models here keep one sign across the whole sweep, positive for Gemma 3 12B and Llama 3.1 8B and negative for Qwen3 8B. The strategy mix moves in all three, by 0.375, 0.388 and 0.278 in total variation against nulls of 0.143, 0.115 and 0.123, and the unconditional rules carry it: AllD swings 0.223

and AllC 0.161 across the six scales, against 0.049 for tit-for-tat and 0.020 for win-stay lose-shift, the last of which does not clear its null at all.

Table S11: The open-weight corpus of the earlier preprint, by model. Three models, six payoff scales, five languages, thirty rounds, 80 agent-games per model × language × scale cell. "Swing" is the largest difference in cooperation rate between two payoff scales, against a permutation null that shuffles the scale label across whole dyads. The persona effect is the cooperation rate of an agent told to be cooperative minus that of an agent told to be selfish, measured within a notation regime, and a dagger marks an estimate whose 95% dyad bootstrap interval excludes zero. "Mix" is the largest total-variation distance between the strategy mixes at two scales. Nothing in this table is comparable with a number from the main text: the horizon is thirty rounds rather than ten, and the read-out uses the corresponding network.

		Coop. shift		Persona effect			Persona shift		Mix shift	
Model	Coop.	swing	null	frac.	unit	large	f.→u.	u.→l.	obs.	null
Gemma 3 12B	0.535	0.316***	0.121	+0.138†	+0.382†	+0.177†	+0.244†	−0.205†	0.375***	0.143
Llama 3.1 8B	0.420	0.198***	0.047	+0.093†	+0.145†	+0.097†	+0.052†	−0.049†	0.388***	0.115
Qwen3 8B	0.474	0.267***	0.095	−0.442†	−0.250†	−0.308†	+0.192†	−0.058	0.278***	0.123

*** $P < 0.001$ over 2,000 dyad-level permutations. Best description by BIC: the three-level notation regime for Llama 3.1 8B and Qwen3 8B, a cubic in $\log_{10} \lambda$ for Gemma 3 12B. Per-strategy swings across the six scales: AllD 0.223, AllC 0.161, tit-for-tat 0.049, win-stay lose-shift 0.020, ambiguous 0.118, against nulls of 0.063, 0.054, 0.038, 0.029 and 0.052.

S3 Supplementary tables

Table S1: The corpus, by model. “Cell n” is the number of agent-games in every model × language × scale cell; the design is balanced, with none missing. The two groups are two sampling densities of one arm and share the base matrix, the horizon and the decoding settings; they differ in whether the round count was disclosed in the prompt, so no level statistic is compared across the rule.

Sweep	Model	Scales	Range	Lang.	Rnds	Dyads	Games	Decis.	Cell n	Coop.
ten scales	Gemini 3.5 Flash-Lite	10	0.01–1000	5	10	2,000	4,000	40,000	80	0.751
ten scales	Gemini 3.1 Flash-Lite Prev.	10	0.01–1000	5	10	2,000	4,000	40,000	80	0.617
ten scales	GPT-5.4 Nano	10	0.01–1000	5	10	2,000	4,000	40,000	80	0.606
three scales	Claude 3.5 Haiku	3	0.1–10	5	10	600	1,200	12,000	80	0.625
three scales	GPT-4o	3	0.1–10	5	10	600	1,200	12,000	80	0.448
three scales	Mistral Large	3	0.1–10	5	10	600	1,200	12,000	80	0.524

Table S2: What the rescaling changes, and what it cannot. The cell values as the prompt template prints them at each scale, with the notation features derived from that string on the left and the game-theoretic invariants on the right. Greed, fear, the dominant action and the equilibrium are constant down every row by construction; only the printed string moves. The row for $\lambda = 0.5$ is the one on which the regime read from the printed cells and the regime read from λ disagree.

Sweep	λ	Printed cells	Regime	Frac.	Glyphs	Digits	Greed	Fear	Dom.	Nash
ten scales	0.01	`0 / 0.02 / 0.06 / 0.1`	fractional	1	3.00	1	0.20	0.40	A	A,A
ten scales	0.1	`0 / 0.2 / 0.6 / 1`	fractional	1	2.00	1	0.20	0.40	A	A,A
ten scales	0.25	`0 / 0.5 / 1.5 / 2.5`	fractional	1	2.50	2	0.20	0.40	A	A,A
ten scales	0.5	`0 / 1 / 3 / 5`	fractional	0	1.00	1	0.20	0.40	A	A,A
ten scales	1	`0 / 2 / 6 / 10`	unit	0	1.25	2	0.20	0.40	A	A,A
ten scales	2	`0 / 4 / 12 / 20`	unit	0	1.50	2	0.20	0.40	A	A,A
ten scales	5	`0 / 10 / 30 / 50`	unit	0	1.75	2	0.20	0.40	A	A,A
ten scales	10	`0 / 20 / 60 / 100`	unit	0	2.00	3	0.20	0.40	A	A,A
ten scales	100	`0 / 200 / 600 / 1000`	large	0	2.75	4	0.20	0.40	A	A,A
ten scales	1000	`0 / 2000 / 6000 / 10000`	large	0	3.50	5	0.20	0.40	A	A,A
three scales	0.1	`0 / 0.2 / 0.6 / 1`	fractional	1	2.00	1	0.20	0.40	A	A,A
three scales	1	`0 / 2 / 6 / 10`	unit	0	1.25	2	0.20	0.40	A	A,A
three scales	10	`0 / 20 / 60 / 100`	unit	0	2.00	3	0.20	0.40	A	A,A

Table S3: The description ladder, model by model. ΔBIC from the best specification for that model, every row fitted to identical data with language and both personas controlled. Bold marks the best specification in each column. No notation specification is best for any of the three models that met ten scales; on three scales the fractional flag and the regime factor are the same specification, and a cubic is collinear with a quadratic, so the rungs that appear to rank are not distinguishable.

	ten scales			three scales		
How λ enters	Gem 3.5	Gem 3.1	Nano	Claude	GPT-4o	Mistral
no λ term	80.9	37.1	220.9	243.3	118.4	**0.0**
linear in $\log_{10}\lambda$	49.1	**0.0**	67.3	61.3	29.3	5.7
quadratic in $\log_{10}\lambda$	56.0	7.6	**0.0**	6.9	7.0	11.9
cubic in $\log_{10}\lambda$	63.8	13.3	5.2	6.9	8.1	11.9
fractional flag	64.2	29.7	121.3	**0.0**	**0.0**	4.9
fractional + glyph count	62.5	8.4	124.7	6.9	7.0	11.9
notation regime	45.7	11.0	129.7	0.0	0.0	4.9
saturated in λ	**0.0**	57.9	10.7	6.9	7.0	11.9

Best description by BIC: Gem 3.5: saturated in λ, Gem 3.1: linear in $\log_{10}\lambda$, Nano: quadratic in $\log_{10}\lambda$, Claude: fractional flag, GPT-4o: fractional flag, Mistral: no λ term.

Table S4: The persona instruction, model by model and scale by scale. The effect is the cooperation rate of an agent told to be cooperative minus that of an agent told to be selfish. A dagger marks an estimate whose 95% dyad bootstrap interval excludes zero. Read across a row: three of the six models hold one sign over the whole sweep and three cross zero, which is why the average over models describes no individual model.

Model	0.01	0.1	0.25	0.5	1	2	5	10	100	1000
the three models that met ten scales										
Gemini 3.5 Flash-Lite	$-0.258^\dagger$	$-0.489^\dagger$	$+0.122^\dagger$	$+0.228^\dagger$	$+0.238^\dagger$	$+0.235^\dagger$	$+0.291^\dagger$	$+0.198^\dagger$	$+0.182^\dagger$	$+0.232^\dagger$
Gemini 3.1 Flash-Lite Prev.	$+0.574^\dagger$	$+0.637^\dagger$	$+0.666^\dagger$	$+0.723^\dagger$	$+0.743^\dagger$	$+0.717^\dagger$	$+0.752^\dagger$	$+0.737^\dagger$	$+0.688^\dagger$	$+0.721^\dagger$
GPT-5.4 Nano	$-0.161^\dagger$	$-0.109^\dagger$	$+0.009$	-0.025	-0.048	$+0.030$	$+0.043$	$+0.050$	-0.025	$+0.098^\dagger$
the three models that met three scales										
Claude 3.5 Haiku	-	$-0.318^\dagger$	-	-	$-0.067^\dagger$	-	-	$-0.057^\dagger$	-	-
GPT-4o	-	$-0.438^\dagger$	-	-	$+0.200^\dagger$	-	-	$+0.305^\dagger$	-	-
Mistral Large	-	$-0.599^\dagger$	-	-	$-0.255^\dagger$	-	-	$-0.152^\dagger$	-	-

Table S5: Strategy-mix shifts on both vocabularies. The largest total-variation distance between the mixes at two payoff scales, against a null that shuffles the scale label across whole dyads. "5-cat." uses the five categories the read-out emits; "named" drops *ambiguous* and renormalises over the four canonical rules, so that a change in how identifiable the play is cannot contribute. "Named share" is the fraction of that model's agent-games the restriction keeps. Every model is significant on both vocabularies, and the restriction raises three of the six estimates rather than lowering them, so none of these shifts is an artefact of play becoming more or less nameable.

Sweep	Model	5-cat.	P	null	named	P	null	Named share
ten scales	pooled	0.134	< 0.001	0.085	0.115	< 0.001	0.076	0.799
ten scales	Gemini 3.5 Flash-Lite	0.338	< 0.001	0.155	0.304	< 0.001	0.150	0.736
ten scales	Gemini 3.1 Flash-Lite Prev.	0.160	0.015	0.145	0.194	< 0.001	0.115	0.754
ten scales	GPT-5.4 Nano	0.312	< 0.001	0.142	0.320	< 0.001	0.135	0.906
three scales	pooled	0.204	< 0.001	0.062	0.196	< 0.001	0.057	0.961
three scales	Claude 3.5 Haiku	0.262	< 0.001	0.115	0.245	< 0.001	0.108	0.953
three scales	GPT-4o	0.325	< 0.001	0.102	0.296	< 0.001	0.098	0.942
three scales	Mistral Large	0.125	0.0055	0.097	0.127	0.0040	0.094	0.988

Table S6: The strategy mix, model by model. Share of that model's agent-games carrying each label at each payoff scale. A dash marks a scale the model never met.

Model	Label	0.01	0.1	0.25	0.5	1	2	5	10	100	1000
the three models that met ten scales											
Gemini 3.5 Flash-Lite	AllC	0.370	0.307	0.405	0.295	0.412	0.357	0.305	0.388	0.297	0.302
	TFT	0.090	0.072	0.107	0.168	0.180	0.180	0.172	0.145	0.170	0.147
	WSLS	0.043	0.098	0.115	0.160	0.223	0.133	0.180	0.198	0.165	0.152
	AllD	0.130	0.233	0.018	0.030	0.020	0.105	0.130	0.075	0.107	0.175
	Ambiguous	0.367	0.290	0.355	0.347	0.165	0.225	0.212	0.195	0.260	0.223
Gemini 3.1 Flash-Lite Prev.	AllC	0.338	0.330	0.338	0.357	0.347	0.297	0.320	0.275	0.275	0.273
	TFT	0.068	0.028	0.033	0.010	0.030	0.040	0.030	0.030	0.072	0.060
	WSLS	0.188	0.182	0.140	0.090	0.117	0.155	0.140	0.128	0.102	0.110
	AllD	0.188	0.215	0.250	0.292	0.255	0.258	0.260	0.318	0.300	0.307
	Ambiguous	0.220	0.245	0.240	0.250	0.250	0.250	0.250	0.250	0.250	0.250
GPT-5.4 Nano	AllC	0.253	0.300	0.380	0.383	0.385	0.425	0.485	0.390	0.455	0.445
	TFT	0.125	0.140	0.117	0.133	0.128	0.138	0.138	0.147	0.150	0.175
	WSLS	0.128	0.138	0.145	0.160	0.175	0.128	0.140	0.115	0.158	0.152
	AllD	0.445	0.312	0.217	0.235	0.253	0.198	0.133	0.228	0.152	0.158
	Ambiguous	0.050	0.110	0.140	0.090	0.060	0.113	0.105	0.120	0.085	0.070
the three models that met three scales											
Claude 3.5 Haiku	AllC	-	0.282	-	-	0.407	-	-	0.438	-	-
	TFT	-	0.207	-	-	0.193	-	-	0.233	-	-
	WSLS	-	0.145	-	-	0.217	-	-	0.175	-	-
	AllD	-	0.350	-	-	0.102	-	-	0.110	-	-
	Ambiguous	-	0.015	-	-	0.080	-	-	0.045	-	-
GPT-4o	AllC	-	0.225	-	-	0.290	-	-	0.255	-	-
	TFT	-	0.087	-	-	0.133	-	-	0.138	-	-
	WSLS	-	0.115	-	-	0.250	-	-	0.250	-	-
	AllD	-	0.568	-	-	0.273	-	-	0.242	-	-
	Ambiguous	-	0.005	-	-	0.055	-	-	0.115	-	-
Mistral Large	AllC	-	0.398	-	-	0.460	-	-	0.425	-	-
	TFT	-	0.020	-	-	0.083	-	-	0.048	-	-
	WSLS	-	0.165	-	-	0.098	-	-	0.168	-	-
	AllD	-	0.403	-	-	0.345	-	-	0.355	-	-
	Ambiguous	-	0.015	-	-	0.015	-	-	0.005	-	-

Table S7: Where each strategy label came from. Share of agent-games resolved at each stage of the read-out. "Provable rule" means exactly one canonical rule reproduces the run of play; "several rules" means more than one does and the run is recorded as ambiguous; the last two are the LSTM's nearest rule, split at the 0.90 posterior floor below which a read-out designed to abstain would decline to answer. None of the four shares is constant across the sweep, so how nameable the play is depends on the payoff scale.

Sweep	Scale	provable rule	several rules	LSTM ≥ 0.90	LSTM < 0.90
ten scales	$\lambda = 0.01$	0.199	0.212	0.362	0.227
ten scales	$\lambda = 0.1$	0.199	0.215	0.333	0.253
ten scales	$\lambda = 0.25$	0.169	0.245	0.348	0.237
ten scales	$\lambda = 0.5$	0.142	0.229	0.396	0.233
ten scales	$\lambda = 1$	0.183	0.158	0.400	0.258
ten scales	$\lambda = 2$	0.146	0.196	0.401	0.258
ten scales	$\lambda = 5$	0.177	0.189	0.390	0.244
ten scales	$\lambda = 10$	0.161	0.188	0.412	0.238
ten scales	$\lambda = 100$	0.167	0.198	0.427	0.208
ten scales	$\lambda = 1000$	0.173	0.181	0.405	0.241
ten scales	all scales	0.172	0.201	0.387	0.240
three scales	$\lambda = 0.1$	0.367	0.012	0.410	0.211
three scales	$\lambda = 1$	0.218	0.050	0.466	0.266
three scales	$\lambda = 10$	0.215	0.055	0.428	0.302
three scales	all scales	0.267	0.039	0.435	0.259

Table S8: Cooperation by prompt language and payoff scale. The FAIRGAME templates do not state the objective identically in every language, so language is confounded with objective framing and the framing each template uses is given alongside it. The five languages do not move in parallel across the sweep.

Language	Framing	0.01	0.1	0.25	0.5	1	2	5	10	100	1000
the three models that met ten scales											
English	minimise	0.712	0.641	0.812	0.783	0.745	0.705	0.665	0.710	0.698	0.662
French	minimise	0.745	0.688	0.778	0.637	0.660	0.608	0.655	0.690	0.582	0.633
Vietnamese	minimise	0.596	0.552	0.684	0.657	0.628	0.645	0.645	0.558	0.670	0.608
Chinese	maximise	0.516	0.635	0.648	0.615	0.595	0.634	0.630	0.641	0.621	0.650
Arabic	maximise	0.549	0.622	0.640	0.706	0.680	0.677	0.723	0.710	0.731	0.632
the three models that met three scales											
English	minimise	-	0.425	-	-	0.588	-	-	0.604	-	-
French	minimise	-	0.373	-	-	0.662	-	-	0.637	-	-
Vietnamese	minimise	-	0.423	-	-	0.535	-	-	0.622	-	-
Chinese	maximise	-	0.424	-	-	0.555	-	-	0.557	-	-
Arabic	maximise	-	0.462	-	-	0.590	-	-	0.527	-	-

Table S9: What one decade of separation buys. The cooperation gap between two scales a decade apart, plus the off-decade pair $\lambda = 0.5$ against $\lambda = 5$, held against the 95th percentile of the permutation null for that model's *full* sweep, which is conservative because a two-level design generates a smaller null. This is the evidence behind the reporting standard, and behind the qualification it carries: two conditions a decade apart detect the effect, but which decade is not a free choice.

Sweep	Model	Contrast	Gap	Null (full sweep)	Clears
ten scales	Gemini 3.5 Flash-Lite	0.01 vs 0.1	0.098	0.082	yes
ten scales	Gemini 3.5 Flash-Lite	0.1 vs 1	0.080	0.082	no
ten scales	Gemini 3.5 Flash-Lite	0.5 vs 5	0.104	0.082	yes
ten scales	Gemini 3.5 Flash-Lite	1 vs 10	0.020	0.082	no
ten scales	Gemini 3.5 Flash-Lite	10 vs 100	0.020	0.082	no
ten scales	Gemini 3.5 Flash-Lite	100 vs 1000	0.052	0.082	no
ten scales	Gemini 3.1 Flash-Lite Prev.	0.01 vs 0.1	0.007	0.077	no
ten scales	Gemini 3.1 Flash-Lite Prev.	0.1 vs 1	0.013	0.077	no
ten scales	Gemini 3.1 Flash-Lite Prev.	0.5 vs 5	0.027	0.077	no
ten scales	Gemini 3.1 Flash-Lite Prev.	1 vs 10	0.037	0.077	no
ten scales	Gemini 3.1 Flash-Lite Prev.	10 vs 100	0.015	0.077	no
ten scales	Gemini 3.1 Flash-Lite Prev.	100 vs 1000	0.007	0.077	no
ten scales	GPT-5.4 Nano	0.01 vs 0.1	0.103	0.080	yes
ten scales	GPT-5.4 Nano	0.1 vs 1	0.035	0.080	no
ten scales	GPT-5.4 Nano	0.5 vs 5	0.082	0.080	yes
ten scales	GPT-5.4 Nano	1 vs 10	0.057	0.080	no
ten scales	GPT-5.4 Nano	10 vs 100	0.030	0.080	no
ten scales	GPT-5.4 Nano	100 vs 1000	0.010	0.080	no
three scales	Claude 3.5 Haiku	0.1 vs 1	0.232	0.058	yes
three scales	Claude 3.5 Haiku	1 vs 10	0.007	0.058	no
three scales	GPT-4o	0.1 vs 1	0.226	0.062	yes
three scales	GPT-4o	1 vs 10	0.008	0.062	no
three scales	Mistral Large	0.1 vs 1	0.036	0.065	no
three scales	Mistral Large	1 vs 10	0.005	0.065	no

Table S10: Cooperation across the game, by notation regime. Round one on its own, then three equal blocks of the remaining rounds. On the three-scale grid the gap between the regimes is open on round one, on which no history exists, and stays open; on the ten-scale grid, where the regime and the magnitude no longer coincide, the three regimes are within a point of each other at every position.

Sweep	Regime	`r1`	`r2-3`	`r4-6`	`r7-10`
ten scales	fractional	0.698	0.694	0.666	0.631
ten scales	unit	0.691	0.713	0.669	0.619
ten scales	large	0.688	0.699	0.660	0.605
three scales	fractional	0.445	0.356	0.410	0.456
three scales	unit	0.618	0.558	0.578	0.603

References

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